

ActInf Textbook Group ~ Cohort 4 ~ Meeting 12

(Chapter 5 part 2)

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all right welcome thanks everyone for joining this is the final recorded session on the first half of cover four it's August 22nd 2023 and we are going to discuss chapter five so before we look at what we might have ahead of us from questions or anything else does anyone want to make any general comments on five I'm just anyone can give a give a thought just since we're on chapter 5 in the neurobiology we talked I think a little bit in the chapter 10 closing discussion for cohort three about some of the topics like what is empirical evidence or what is any experimental setting have to do with free energy principle octave inference and and we reflect a little bit on this recent paper that like with isomura at all that like broke through kind of another level not necessarily even from an epistemic perspective oh I guess in terms of people's awareness it was epistemic in that sense then I was thinking about this was in 2018. I mean it's the same title I know that the work has proceeded a lot since then so that's not the same they rehash the same but in some ways because there is a difference between map and territory between like models of X and X in a way there's like this eternal Gap that just continues to swallow up experimental results um much to the Chagrin of people who think that just one more experiment is gonna gonna Bridge or map the Gap and that's kind of like what some of us are told as graduate students a lot of graduate school has to do with like creating a data set and people get into the zone of thinking that that one data set is going to be addressing the question even a narrow one and of course like if it's not helpful then let's not waste our resources to do that but at the same time people were um saying that the relationship between gene expression and behavior was resolved in the 90s and it's the same language that comes out today and then even in this kind of fractal free energy way it's like what were we looking for in the petri dish again Ollie yeah that was exactly uh what I was uh gonna say because uh in fact I would love to see in newer editions of the textbook to uh to see some of the new directions and what some people call radical biology uh to be incorporated much more uh that than we see uh currently in the textbook because uh actually the The Works of scientists such

as Michael 11 and even before that I mean Varela Montana and copra they show some radically new approaches to see see biology and and even life itself through I mean the conceptions of autopilotoisis and allopoiesis and other uh related terms but I believe that closer incorporation of these Concepts uh into the textbook can help to see the bigger picture which connects Bayesian mechanics as an overarching theory of every quote-unquote thing to active inference because the way it stands now uh and in particular the way chapter 5 stands now there seems uh quite a big gap between uh what Bayesian mechanics represents as a theory of every thing and it's neurobiological representation or I don't know neurobiological evidence or whatever but if there was a I mean more material on these new directions of radical biology I believe these gaps can begin to be filled and the bigger picture can be um can finally emerge as a kind of continuous picture with without any significant um and possibly confusing gaps in between anyone else want to give a thought on just one speculation here the map territory Theory practice the sort of dualistic dialectic is well known questions like what does math have to do with Biology and what is statistics have to do with something like a system of Interest so um that's kind of the classic we're dealing with another theoretical distinction within the domain of theory were making a distinction not not a uh homogenization but a distinction between the more theoretical physics axiomatic View and the lens or tradition or method or approach or process Theory described by active inference that level of um disentanglement of kind of like the platonic core from the more manifest models in theory but theory in practice and that's like exactly where active sits so I think in the live chat for the Symposium session duvid and I were kind of going back and forth and his contention was like well octave inference has been like really um useful in the machine learning and artificial intelligence space but it hasn't it hasn't made any contributions to to neuroscience and then I mean there's so many angles to take on that point but like one is um is that it hasn't made many contributions relative to what you think another theory has done or is it it hasn't made enough contributions relative to what your preferences for what it could have done by this time are that that's um one angle but then the other angle that kind of presented itself was like like are we looking for experimental validation of the free energy principle or are we looking for people who want to apply active inference and that's the experimental validation I mean what if it was experimental validation of sum of squares linear regression on electronic Healthcare record data be like talk about a non-sequiturer decade after decade experimental validation of linear regression on rainfall and tree growth data how many decades before you just say we used a linear regression and if you think we could have used it better then tell us how or if you

have another model other than a regression let's see it but that mobilizes the discussion in like a very radical way Ali uh yes I I also don't think uh that active inference or fep uh can be considered as a kind of Niche theoretical thinking anymore because uh recently I I'm more and more often I come across some articles or books that don't mention active inference or fep at all or I don't know any work related to active inference but the ideas presented in these works are very very similar to what active inference to active inferences assertions so it it looks like there's the box and active inference or fep fits right into I mean it's a theory that is proposed uh uh I mean at the right time uh and uh for the Right audience I believe because uh if it were proposed it was proposed about I don't know three decades ago uh maybe uh or probably it could go to into Oblivion without uh people being aware of any of its significance at all but uh as I say said it seems like the present time uh is kind of uh fertile for these kinds of not radical but a bit out of the box thinking that uh would that can potentially help to reshape our understanding of life in its broader sense so yeah again its current status seems to me like it's beginning to be uh mainstream or at least in the early phases of a mainstream theory is truly evident from some recent work in this area yeah and it's so fun to see the memes in the narrative sharpen like when I was just walking around a few minutes ago I was thinking okay reward maximization or Precision optimization on getting more reward same thing but also surprise bounding is more General than reward but you could make a peer reward model we see that in in the textbook so it's like what's your first principle reward surprise bounding or something else or nothing or you haven't thought about it it's all it's all fine but which one of them is it or you or do you want to pop out a level and even have like a different way of approaching the question so it's not like presenting a closed box it's like a more dynamical frame which is largely potentially even overwhelmingly congruent with simply modern statistical physics and again then do you want to say well I have an alternate statistical physics branch or I have higher Visions for what statistical physics could do than what it is doing it's like congratulations then you're a researcher now it still may be decades or never until you make a contribution but that's exactly the path of a researcher seeing an area and then thinking that it can go beyond and like well I don't have that alternative proposal for Niche construction yet but I want to be in the game to be there to recognize when it does happen or I want to make something happen eventually merely waiting at the starting line comparing Frameworks to on described alternatives we barely even have the vocabulary to describe some of these um conversational patterns and now we have more flexibility to like pop out and pull back to the meta science but again it's like okay so are you saying we can't use math to use to do in the

context of Neuroscience and some people may hold that perspective and that may be the conversation to have at that time but once the um or or somebody may think that the engineering tools like Bert brought this up many times in the symposium that machine learning benefits greatly from Decades of development on toolboxes and uh in in his section he was like yeah this is like the optimistic positive message is within months to years and Carl also alluded to it in the round table like we will have closer to push button Solutions so I think that really contextualizes a lot of what we're doing and learning when a lot of these formalities are going to get just like simply baked in and people will be engaging at at more applied layers but all in the context of chapter five where a variety of empirical data are being brought into play we're talking about real Anatomy talking about real reflexes neurotransmitters real design patterns in the nervous system in the body so does anyone want to give a thought or ask any chapter five related questions all right let's just see if we can up just see what comes through okay um schematic in figure five one shows layer 5 of the cortex projects to spinal pyramidal neurons this can be interpreted as a prediction spinal motor neuron and spinal pyramidal neurons are sometimes used interchangeably SP okay anyone have a preference for which one of these questions to jump into I would be curious about the the question regarding how do habits and goals interact um I found this chapter interesting in part just because the introduction of habits or like the e variable I suppose yeah well let's start with what is the goal we're gonna have to reach outside of the active inference ontology to bring it in because we don't need goals although at the same time the term is used prolificately let's see how it's used in five this is a mixed mixed dialect e and G are habitual and goal-directed drives that's a very interesting sense of goal like in the um football soccer setting the goal is like a fixed external objective like a milestone but a culminating milestone but how is how is it being used here if by goal we mean something that g sharpens well in that case well first let's look at what e does e is the prior on policies so it reflects before the observation comes in before updating has happened what are the prior probabilities of certain actions being taken and then that observation may not influence e so e kind of gets passed through and we see that in the basal ganglia second example in the chapter um or or it gets sharpened so if we see expected free energy minimization as being goal directed certainly it's not sufficient to say that goals are pragmatic only because if we take everything that g encompasses like that better policies as evaluated by G are more goal directed and inferior policies are less goal directed then up in the category of more goal aligned like let's just say the two most goal directed policies they may radically differ in their composition because one may be heavily loaded on pragmatic value

homeostasis bringing observations into alignment with preferences and all that and the other policy might have a good expected free energy because of its high epistemic value and so it seems like at least just locally here though probably elsewhere in the book they're talking about goals in just this kind of broader cybernetic goal-oriented sense here we see the goal is like the process and the goal is this imperative or maybe there's other ways to read it too but it's kind of like G is it's like if you were trying to drive somewhere there's like a location you're trying to get there with a location and a Time you could you could also think about that as Precision in space and time which decisions are going to be more goal oriented well if the goal is just the location on the soccer pitch then the most goal-oriented policy is the one that's like locally or globally taking you closest to the location on the soccer field so that's a situation where like you're like yeah there's time but really like let's just pursue this the space or location as fast as possible we'll get there when we get there but then if you say well we're going to hold up both of these features in this driving example the the space and the time but in G the epistemic and pragmatic and we're going to say well actually the joints optimization of those two is what we're going to call our goal that's a little different sense of goal but it is used here and it's specifically used to talk about how habits and goals differ namely where habits are reflecting the prior probability on taking different actions kind of the pass-through model where observations don't influence what actions are likely to be taken those still actions may be taken at in equal frequencies whereas goal orientation means an increasing role for expected free energy in sharpening that policy prior into policy posterior Ali oh sorry uh well yeah another relevant uh remark here is uh about this uh you see uh there's this kind of um telephobia among biologists and uh they usually try to uh avoid uh using the word purpose instead of go because they believe that although biological agency implies a kind of a goal-directed autonomy uh it doesn't necessarily require for those agents to be um to have any kind of purposiveness as uh then it would and argue but the problem here I mean the distinguishing between these two concepts uh is a kind of I mean as I said it probably emerges from this um telephobia among biologists because uh any talk of purposiveness can be seen as a provoking teleology which is something that scientific culture has been very averse from many centuries ago but if we somehow Define goals or purposes as a kind of mechanistic lawfulness to apparently purposive Behavior then this dilemma between these two concepts would probably be resolved because in this case we have basically defined a goal in terms of purposiveness so they're not actually separate terms and they co-constitute each other uh of course the same kind of phobia was evident probably much more it was a bit more prominent a couple of

decades ago but it was also evident in the for for the concept of function but but it seems like this phobia has been diminished in recent years but nevertheless this teleophobic refusal to accept purposes as real factors in biology Still Remains but active inference and some of these recent progresses in biology as I said the works of Michael Levin and others are ways to somehow dismantle this dichotomy and this uh teleophobia and as David cracker would call to see complex systems as a kind of teleonomic matter so uh it doesn't necessarily something that would require us to think about any uh I don't know Supernatural or Supernatural entities or transcendental entities to get into the picture we can talk about purely naturalistic and mechanistic teleology as well using these goals and purposes in this kind of interrelated terms yeah great comments one more piece on that from like the so-called argument by Design which is actually played a very interesting role arising most famously in the the watchmaker analogy which is like you're walking on putting aside the big meta which is like what is the utility of any thought experiment or any like little narrative walkway that somebody provides you it just is like if you look at a watch you're going to infer intention of a Creator and so then the argument sometimes plays out like we'll look at all the small bones in the ear or look at the way that the retina is um then the design argument became actually one of the design and Engineering became some of the key pieces of the modern evolutionary synthesis in evolutionary biology but as Ali alludes to there always was this kind of teleophobia as Mike Levin calls it sometimes or just like this awkwardness around like well are we going to appeal to just all functions just as they are so that's like a sort of like functional teleology just or or does one go the kind of tehar De chardine Omega Point Bobby azarian angle like that is the account and where the radical biology in a way speaks that or cuts the knot is like the the hidden fallacy in the watchmaker thus is like it tries to transfer the analogy of a top-down function [Music] bestowments whereas if we have self-creating or co-creating or embedded arisel then we can say yeah like the Circadian circuit arose through Evolution and development as a bottom up self-created in relation mechanism for just that telling time so at the level of linguistic course training it makes sense to say that the circadian clock keeps time just speaking super broadly here like a watch keeps time but there's a critical difference which is that the watch was like an engineering artifact as the top-down projection of a cognitive system whereas some kind of like internal organic representational system like a circadian clock is actually like a bottom-up construction so like there just isn't an issue with describing or understanding and it's just an English word anyway once we really start to get into the specifics with agency and habit a lot of this may just dissipate and in the more recent multi-

scale formalizations the habits can include like like the phylogenetic scale or the Habit could refer to just what is passed through moment to moment in the organism's life or you can imagine this as being like a intergenerational transfer and a habit or a a developmental tendency for a habit is also able to be inherited Andrew any thoughts on that or wherever anyone else wants to go no I don't have any any deeper thoughts on on that I suppose uh I guess I'm more interested in the sort of computational aspect of it as far as message passing goes rather than um I mean I find all of the the kind of the more abstract philosophical side of of discussing what goals are very interesting and in certain ways very important um when putting together a framework but I guess here it's just um I just found it interesting like oh now we have habits in addition to G which I previously considered to be like just expected free energy um but maybe I have my terminology mixed up here is this G not the same as um oh you you're totally right and I think that that's just it's a little um coincidental alliteration and G never stands for goals it's just a total typographical spandrel so then the question becomes what is the goal concept broad enough to Encompass what it is that g does and that's clearly not the same thing as just like the mile marker on the side of the road but then a little bit more on the um computational side e primes the pump for the procedure that g does like the shape of the variable e is the same shape that gets taken into G the cells in the in the variable e are what get sharpened by G as it iterates through every policy and calculates its expected free energy so e and G they both are engaging with this policy variable which is just a list of all the possible policies so all the affordances basically raised to the power of how many time steps that you want to play it out for G sharpens what e provides or modifies what e provides and the gamma the temperature parameter on on decision making here associated with dopamine neuromodulation modulating that trade-off between the the E pass forward in the indirect pathway so it's kind of type one and then the policy updating driven by G sure no thanks for kind of giving more explanation on that it's um so I find that very interesting then because at some point in here it refers to like uh um sorry I can't find it immediately but the use of the word context um so so um it's as if e so those are your priors over policies and so you're lacking something like context of the given the agent is lacking context in a given moment I suppose such that one might like rely upon um their habits to deal with the scenario whereas G is in fact taking into account not only e but context in the sense of oh uh the agent is taking in observations um it's Computing expected free energy um it I I guess I'm just putting everything together as far as I understand it so it's e Prime you start with e you enter into some context with a variety of observations or otherwise from there you begin to compute G and

then from there you can start to actually score potential policies to be taken with some of them deemed to be more minimizing a free energy given the context and so on versus Eve which seems to be more or less static through the whole computation dissolve that track yeah great points so definitely when you set it up you could make it so that E is fixed or that you can make it so the E counts like earns that's one simple learning technique but but it's a great connection with context so like let's just say that we are um playing cards and then it's like the main idea is we have beliefs about how we act in particular contexts you know I we're playing um 21 Blackjack and I stay 80 of the time just habitually so the two cards are dealt face down habit is before the context is recognized or uh paying no attention to context you take the the 80 stay and then context we could translate as basically like the sensory observations and and the ensuring sense making so now you look at the cards and then you're going to sharpen through G again that doesn't even mean it's better than playing on reflex it doesn't mean it's better than playing 50 50 it doesn't mean you're going to win it doesn't mean like all these like kind of outcome oriented possibilities in the world it is the case like if one lets the card that they saw influence their decision making we would model that as the generative model using expected free energy based policy evaluation planning as inference even if it's just one time step planning that's how we understand the situation where a person who stays eighty percent of the time looks at the cards and makes a decision that differs from that or it could be who else is at the table and then so then they they want to make a certain move not because it aligns their pragmatic value with winning the pot but that it gains them information about who they're playing with and their strategies which speaks this kind of thinking through other Minds topic and also just the need to build specific generative models because how could we talk about what is an expected free energy minimizing decision in the context of a of a blackjack game unless you've delineated like what is even being considered but now think about how far people go beyond the Blackjack game talking about what does and doesn't reduce expected free energy so thanks again for that that was very clarifying and uh yeah so suddenly the phrase is um model free versus model based systems in this context makes a lot more sense yeah model free just kind of flailing but that may be in effect to the flail and it may be an effective way to go or model based is where you get more explicit structuring um okay this one's a freebie popcorn in the chat or however what other neural systems are interesting to model other than the three that are in chapter five what's another mammalian neural region or non-mammalian system that would be fun to model actually I would very love to see active inference applied for modeling emotion circuitry and in fact it's an ongoing going projects of

mine but it proves much more complicated than I previously imagined as uh well some preliminary Works have been done by people like Ryan Smith and others but there still needs I mean lots of work for these kind this kind of modeling to to be as mature and sophisticated as some other areas that active inferences has been applied to but uh for me personally uh it's um it's really interesting Prospect to see how modeling uh emotions are a dream and even modeling emotion itself using active inference terminology and Concepts would reveal some deep mysteries about emotion and its relation to cognition and other mental faculties so yeah that's one area of Interest rule a few things that um Dr NL Lemke said today in the round table that that stakumi one was like about this co-presentation I won't even call it attention or a paradox between accepting higher power and taking agency in one's own life but she related to her like context in the Psychiatry and addiction setting and then also it came up later um with precision and uh monitoring and the increasing ability for monitoring of our biometrics and like just the bigger picture on how like maybe knowing your heart rate at all times maybe it spirals off in all these different ways like now you have too much Precision on heart rate you're paying too much attention to it or there was just a lot of interesting pieces there and then also there was a discussion about like limitations of the brain as a predictive system or of modeling the brain as a protective system and it yet again another place where action comes in to like save the day slash slam dunk because it's like well yeah when it's a prediction about what you're about to do it's only a prediction for so long and then it happens so it's like a lot of these like prediction like they're they're often implicitly in this kind of like um archaic dark room setting like where actions can't be taken and it's just a perceptual question but we know that perception plays a actually plays a role in perception itself through the isocade and through the example at the beginning of the textbook where like you can't determine the texture of a fabric without moving your finger so it's like well it's not demeaning to say that it's predictive when it gets implemented a split second later but all but so much of the predictive discussion focuses just on like on like you know Predictive Analytics predictive forecasting what's going to be the most popular song next year and that's a very removed way of considering the role of action and inference but yeah a lot of neural systems to okay just kind of a related question what point does oh Ollie go for it uh yeah I just wanted to make a maybe a sideways remarks but there's even this recent Trend toward um the the probabilistic aesthetic theory of art so and especially as applied to avant-garde or Contemporary Art so and then they call it predictive Arts uh so uh one Pioneer in this area um Conrad uh woynovsky which is uh who also has an upcoming uh book um I think it will be published in February 2024 but yeah

as you said these kinds of probabilistic modes of um sensing and making sense of reality uh has it started to being applied to even Aesthetics and I mean Contemporary Art criticism as well well yeah I remember this this paper this was I was in the um the the buy it based on the title phase of my graduate school access to free printing and I just remember thinking this was like the funniest and boldest title because there's like several sentences and it has all these weird Illusions in it but um this uh and I think active goes beyond it but on the left side here is the time scales like from like the the 100 millisecond you know visual region this and that and the shorter term memory and okay we're at an exhibit so there's the context window of like what we've seen from the exhibition and then it even expands out to like let's see how slow it gets in the very slowest very very slow like lifelong Mastery I think or something transformation 60 plus seconds but then but then the transformative capacity and and and Bert talked about in the Symposium with like the need for longer term lower resolution estimates and how a key engineering question right now is being able to adjust basically the computational resources allocated to different kinds of estimates so I gave an example of like if you're going to drive three quarters of the way around a roundabout you don't need to know like what centimeter it's going to be at you just want to know what Lane you're going to be in but the next centimeter might matter for the next centimeter and so being able to like dynamically tune or develop hierarchical structures or other kinds of inference uh architectures where we're getting a lot of like bang for the buck because we're using our CPU cycles and other resources But ultimately attention and Trust on we're not calculating 12 decibel points for something that could have just been left at one and vice versa where does reasoning sit in five five I think it depends what you Define reasoning as but looks like in some previous discussions we've talked about like reasoning as unconscious inference and just depends on what is meant by reasoning is it any kind of Reckoning or is it specifically like a consciously aware one but that's that's a good question is fep applicable to any system let's vote The Perennial question coming to flip a coin I'm gonna flip a coin it's gonna be a real coin flip I'm gonna use the real entropy of my room it was not a coin with heads or tails so it was not a conclusive answer but I hope that in future figure out more generalist tools to interact with Bayesian mechanics I think I remember this question being asked and who asked it and the mechanisms in the brain at the uh on this topic actually the uh the first presentation in the symposium Andre bastos this was very much like welcome to the Neuroscience conference like not not grandstanding at all with a theoretical connection just like this is how it's being visually processed here's how we're doing more and more measurements here's how theories

guiding our experiments and it did come up to this like um multi-band Alpha Beta and then the gamma like the differential which we talked about I think in one or two times ago differential frequencies cohort three um sorry Daniel there was this question I think the one down the last one you okay I talked about about uh is there any alternative representation for something like that yeah are there alternative visualizations that better uh so I think we have an additional answer to this question now uh because of chantal's work and string diagrams so yeah I I believe Jana made this image and then I know that Bert brookers has done a ton of work in like alternative Visual and and tactile metaphors and then on the more formal side you're absolutely right that the string diagram may even or some of the representations that Sanjeev is developing in the fundamentals work certainly we're seeing things that are like familiar but also different great works cohort for um that's our final recorded discussion next week we'll have um feedback and talk a little bit then we'll take a few weeks off get the gang back together for cohort for part two and for cohort five and we'll um talk more next week about the timing for everything and see what works for people so thank you all till next time thanks everyone bye thank you